PM SHRI SCHOOL

Jawahar Navodaya Vidyalaya

Monacherra, Hailakandi

Jawahar Navodaya Vidyalaya

Theme — *Science & Technology for Sustainable Future*

Sub-Theme — *Emerging Technologies*

Name — *Shrayan Das*

Class — *XII - Science*

Subject — *Information Technology (IT)*

Guide's Name — *Mr. Rajiv Anand (PGT CS)*

Session — *2026–2027*

EXTINGO

An Autonomous Sensor-Fused Fire Detection & Suppression Robot

Synopsis

Extingo is an Arduino Uno-based robot for autonomous fire detection and suppression. A pan servo rotates its sensor head through North, East, South and West, while a perpendicular tilt servo sweeps elevation at each stop; an IR flame sensor mounted parallel to the tilt arm scans continuously along this path. An I2C OLED confirms “Extingo – Ready” at startup and thereafter refreshes live sensor readings every 3 seconds.

The IR sensor, as the highest-priority cue, triggers the response the instant flame is detected; the gimbal holds its position, and a DHT22 heat reading and MQ2 smoke reading are logged alongside as corroborating, lower-priority evidence. The Uno then reads a GPS module for coordinates and sends an SMS — stating the detected direction and location — before placing an alert call to the owner.

An exhaust fan (via L298N) and an ISD1820 siren activate and loop until the IR sensor clears, while a PIR motion sensor decides pump pressure — low if a person is nearby, high if the area is empty. Independently, a DHT22-based rate-of-rise watchdog flags sudden temperature spikes with an early SMS, runs the exhaust, and sends an “all clear” confirmation once heat stabilizes.

thank you